1. Prove " Perceptron learning algorithm — processes of perceptron learning algorithm — step3. match predicted output P_y & output y(ground truth) : [step3 is only for single perceptron, not fit for neural network] ":

⇒ set weighted sum function $z = w_0 \cdot x_0 + w_1 \cdot x_1 \cdots \cdots w_n \cdot x_n$

⇒ compute gradient descent of $z(w_0, w_1, \ldots w_n) = -\nabla z = -\begin{bmatrix} \frac{\partial z}{\partial w_0} \\ \frac{\partial z}{\partial w_1} \\ \vdots \\ \frac{\partial z}{\partial w_n} \end{bmatrix}$

⇒ use gradient descent method :

$$\vec{W}' = \vec{W} + \eta \cdot (-\nabla z) \longrightarrow \begin{bmatrix} w_0' \\ w_1' \\ \vdots \\ w_n' \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} \frac{\partial z}{\partial w_0} \\ \frac{\partial z}{\partial w_1} \\ \vdots \\ \frac{\partial z}{\partial w_n} \end{bmatrix} \right) \longrightarrow \begin{bmatrix} w_0' \\ w_1' \\ \vdots \\ w_n' \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \right) \longrightarrow \left(\begin{bmatrix} w_0' = w_0' - \eta \cdot x_0 \\ w_1' = w_1' - \eta \cdot x_1 \\ \vdots \\ w_n' = w_n' - \eta \cdot x_n \end{bmatrix} \right)$$

⇒ $\left(\begin{bmatrix} w_0' = w_0' - \eta \cdot x_0 \\ w_1' = w_1' - \eta \cdot x_1 \\ \vdots \\ w_n' = w_n' - \eta \cdot x_n \end{bmatrix} \right)$ is the same as

step3. match predicted output P_y & output y(ground truth) :

```
# if "( p_y <0 ) != ( y = +1 ) " = " ( p_y = -1 ) != ( y = +1 ) "
 ,then " add each w[i] by η * x[i]" = " w[i] += ( y * η * x[i] ), y = +1"

# if " ( p_y >0 ) != ( y = -1 )" = " ( p_y = +1 ) != ( y = -1 ) "
 ,then " subtract each w[i] by η * x[i]" = " w[i] += ( y * η * x[i] ), y = -1"
```

2. Gradient Descent Method vs. Mini-Batch Gradient Descent Method vs. Stochastic Gradient Descent Method (SGD)

A. gradient descent method : compute the mean value of gradient descent for entire training examples, and then updating weights.

B. mini-batch gradient descent method : compute the mean value of gradient descent for a mini-batch (a small set of entire training examples), and then updating weights.

C. stochastic gradient descent method : compute the value of gradient descent for a single training example, and updating weights.